NOTES

Paleomagnetism of Eocene igneous intrusions in the Valley and Ridge Province, Virginia and West Virginia

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Paleomagnetic measurements from 11 small dikes and plugs in the Valley and Ridge Province of Virginia and West Virginia yield a mean magnetic direction of $D = 176.3°$, $I = -60.9°$ after AF cleaning. The geomagnetic pole position calculated from this direction lies at 85.5°N, 243.7°E. The proximity of this pole to others from North American rocks of Eocene age is consistent with radiometric age determinations. This provides further evidence that North America has undergone little apparent polar wander since early Tertiary time, and that the Appalachian Eocene activity covered a wider area than previously realized.

Les mesures paléomagnétiques de 11 petits dykes et bysmalites dans la province de Valley and Ridge en Virginie et en Virginie occidentale donnent une direction magnétique moyenne de $D = 176{,}3°$, $I = -60{,}9°$ après nettoyage dans un champ alternatif. La position du pôle géomagnétique calculée à partir de cette direction se trouve à 85,5°N, 243,7°E. La proximité de ce pôle avec d'autres pôles des roches de l'Eocène d'Amérique du Nord est compatible avec les déterminations d'âge radiométrique. Ceci fournit des preuves additionnelles que l'Amérique du Nord a subi peu de cheminement polaire apparent depuis le début du Tertiaire et que l'activité des Appalaches au cours de l'Eocène a couvert une surface plus grande qu'on ne l'avait réalisé jusqu'à présent.

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Introduction

The occurrence of minor igneous intrusions in the Valley and Ridge province of Virginia and West Virginia has long been known (Darton and Diller 1890; Darton and Keith 1898). The intrusions are mostly northwest-striking dikes, cutting tightly folded limestone, shales, and sandstones of Ordovician to Devonian age (Fig. 1). Although the intrusions of Pendleton County, West Virginia are predominately basalts, a wide variety of igneous rock types are represented; in Highland County, Virginia, andesites are more numerous. On the basis of petrographic resemblances to other eastern North American dikes, Garner (1956) assumed the basalts to be of Triassic age. The andesites had been postulated to be related to the Devonian Tioga bentonite. It came as a surprise, therefore, when Fullagar and Bottino (1969) determined an Eocene age for biotites from the andesite.

Several geophysical anomalies are located near the Eocene rocks and may be causally related to them (Dennison and Johnson 1971; Bollinger and Gilbert 1974). These include the most extensive concentration of thermal springs in eastern North America, an 'earthquake shadow zone' with reduced surface disturbances associated with the 1886 Charleston, South Carolina earthquake, a low number of historical earthquakes in the region, and a Bouguer gravity anomaly less than −80 mGal. Dennison and Johnson (1971) suggested a still-cooling granitic pluton as the cause of the anomalies and as a possible parent magma of the andesites. Hall (1974), however, using feldspar structure and composition, proposed that the andesites could have been derived by fractional crystallization of a basic liquid represented by the basaltic dikes in the area.

Løvlie and Opdyke (1974) provided further evidence for a temporal relationship between the basalts and the andesites in their paleomagnetic investigation. In addition to obtaining a magnetic direction consistent with an Eocene age for four of the andesites, they found a similar direction in the one mafic unit they sampled.

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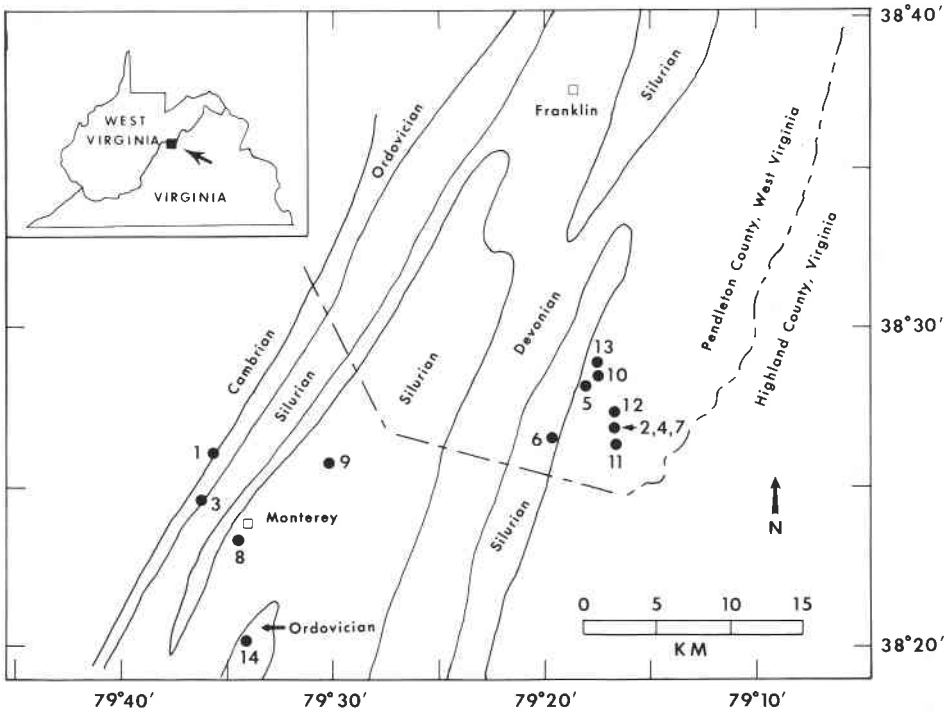


FIG. 1. Generalized map of sampling area with site locations (after Garner 1956; Press and Siever 1974, p. 903).

Our paleomagnetic investigation was undertaken with two purposes. The first was to concentrate on the basalts in Pendleton County to determine if they were all of Eocene age or if some Triassic units could be found. The second was to provide additional data for establishing an Eocene pole position for eastern North America. Independently, some of the same units were sampled for K–Ar age measurements (Core *et al.* 1974). Detailed geologic descriptions of the study area are given by Kettren (1971) for Highland County and by Garner (1956) for Pendleton County.

Sampling and Measurement

Although over 90 intrusions are known in the region only 14 were sampled, owing to the weathered nature of some outcrops and the difficulty in locating others in thick brush. Our sites 1 and 3 are Løvlie and Opdyke's (1974) sites 3 and 8, respectively. At least six oriented cores were drilled with a gasoline-powered drill at each site. Cores were oriented with a compass, sightings being taken along the outcrop to insure that there was no effect on the compass from the dike. After trimming to standard size (2.5 × 2.28 cm), remanent magnetization was measured on either a Schonstedt SSM-1A or Digico complete results spinner magnetometer. Duplicate measurements were made for several samples on both instruments to assure that variations between them were within acceptable limits. Secondary magnetic components were removed by progressive demagnetization in alternating fields using a Schonstedt GSD-1 demagnetizer.

Our procedure was to select one or two pilot samples from each site and demagnetize these at intervals of 50 Oe to 500 Oe, followed by steps of 100 Oe to 1000 Oe. When plots on standard stable end point diagrams (Fig. 2) showed that secondary components had been removed, the remaining samples were demagnetized in the appropriate field. The decrease in intensity with higher demagnetizing fields indicates magnetite is the carrier of the remanence, as was demonstrated by Løvlie and Opdyke (1974). The clustering of directions within a site following cleaning is shown in Fig. 3.

Three of the sites did not yield reliable magnetic directions and were omitted from further calculations. Site 12 was rejected since stable end-point diagrams showed that a single magnetic component could not be isolated. Site 13, a mica-pyroxenite at Garner's (1956) Station 6, contained strong viscous components of magnetization. A Jurassic K–Ar age has been determined for this dike (J. Sutter,

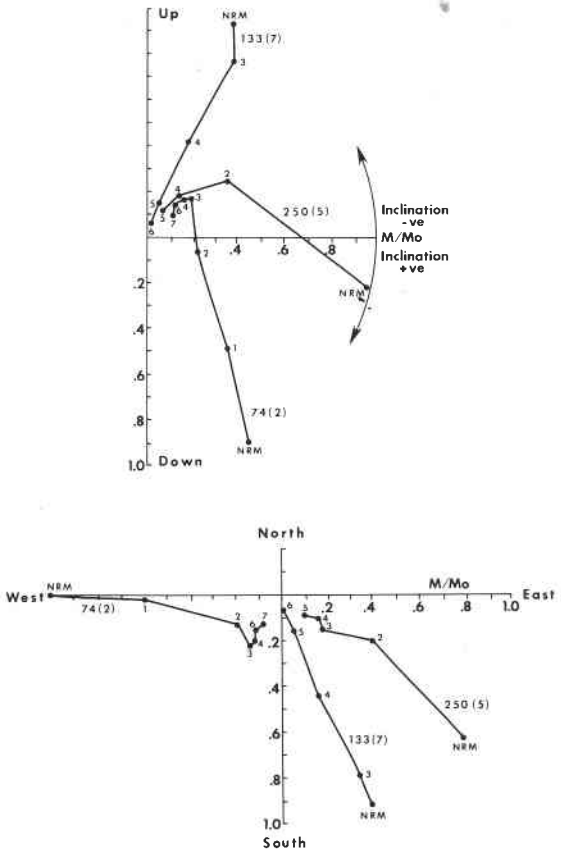


FIG. 2. Stable end-point diagrams showing changes in direction and intensity with AF demagnetization. Changes in the vertical plane are shown above, changes in the horizontal plane are below. M/Mo refers to normalized intensity and is measured outward from the origin. Site numbers are in parentheses following sample numbers. Demagnetization stages are numbered as follows: 1–25 Oe, 2–50 Oe, 3–127 Oe, 4–200 Oe, 5–300 Oe, 6–400 Oe, 7–500 Oe.

personal communication, 1976) and it may be related to other Jurassic alkalic rocks to the east (Zartman *et al.* 1967). Samples from site 14, an amygdaloid basalt at Sounding Knob on Jack Mountain, a local, treeless topographic high, were probably affected by lightning strikes. This site did not give α_{95} values (radii of circles of confidence) less than 45°.

Discussion of Results

The paleomagnetic data from the eleven acceptable sites are summarized in Table 1. Included are five K–Ar ages measured at Ohio State University (J. Sutter, personal communication, 1976) and one determined by us at the Georgia Institute of Technology (site 10). A stereogram of site mean magnetic directions is shown in Fig. 4. All of the sites are reversely magnetized. Ten are closely grouped about the mean magnetic direction of 176.3° declination, −60.9° inclination. The geomagnetic pole position calculated from this direction is 85.5° north latitude, 243.7° east longitude.

The directions we measured for sites 1 and 3 differ from the directions reported by Løvlie and Opdyke (1974) by 18° and 16° respectively. Both their measurements and ours had very small circles of confidence so that the differences are statistically significant. While in the case of site 1 the possibility exists that different dikes were sampled, we have no explanation for the other discrepancy. It should be noted, however, that the directions of Løvlie and Opdyke would also fall in the cluster of directions shown in Fig. 4.

The only magnetic direction that is clearly significantly different from the mean direction is that from site 10. This direction yields a

TABLE 1. Site mean directions and K–Ar ages of igneous units

Site	Rock type	n	$D°$	$I°$	K	α_{95}	Demag. (Oe)	Int.	K–Ar age (Ma)
1	biotite andesite	9	203.1	−61.9	199.9	3.6	250	0.014–0.124	46.5
2	olivine basalt	5	214.8	−62.7	24.5	15.8	250	1.42–2.17	
3	volcanic breccia	12	181.9	−77.1	129.8	3.8	250	2.50–4.28	
4	basalt	7	173.5	−70.3	99.5	6.0	100	1.08–7.50	44.6
5	basalt	12	179.5	−59.3	10.7	13.8	350	5.10–13.66	
6	basalt	6	186.1	−61.4	64.6	8.3	200	0.11–1.51	
7	basalt	10	153.2	−70.3	129.9	4.2	200	1.30–8.79	46.1
8	volcanic breccia	6	179.5	−55.0	21.6	14.8	200	1.66–5.86	
9	diatreme	4	172.3	−51.2	80.8	10.2	250	0.071–0.466	
10	olivine basalt	9	169.0	−22.1	22.5	11.9	175	0.70–2.02	45.9
11	albitite	6	133.2	−59.4	33.4	10.5	1000	0.010–0.095	42.0
Mean		11	176.3	−60.9	21.2	10.1			
Geomagnetic pole position: 85.5°N, 243.7°E dp = 12, dm = 16									

NOTES: n = number of samples; $D°$ = declination; $I°$ = inclination; K, α_{95} = dispersion statistics; Int. = range of NRM intensities in emu × 10^{-3}/cm^3.

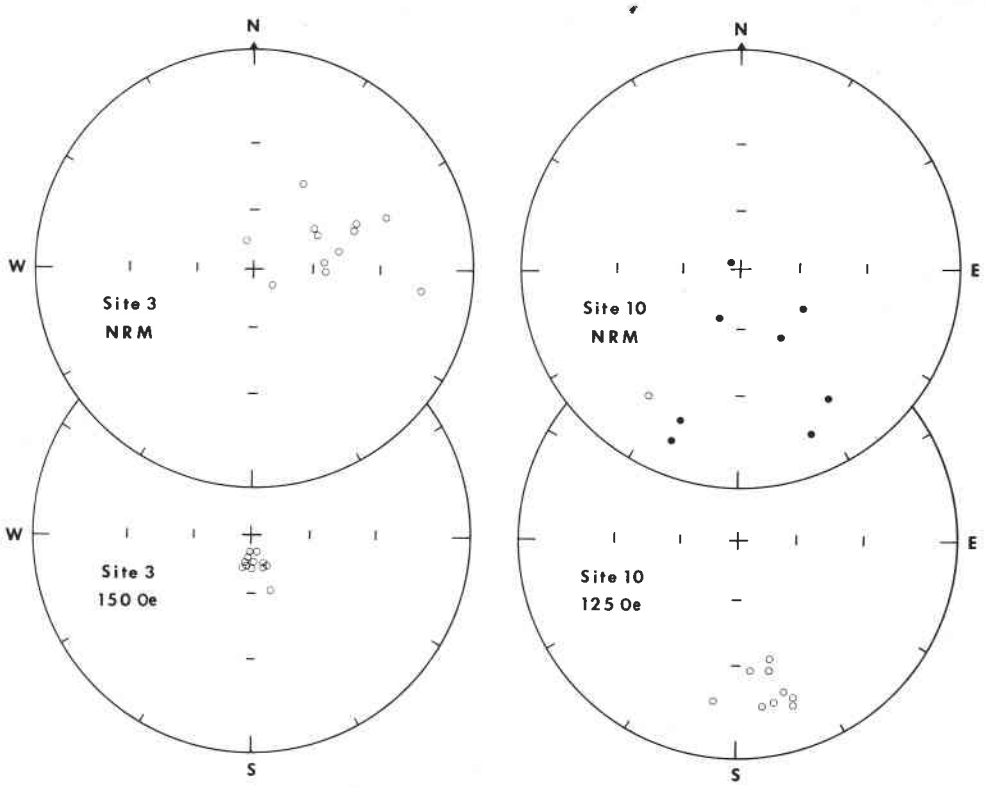


FIG. 3. Stereograms of natural remanent magnetization directions (top) and directions after AF cleaning (below). Negative inclinations are represented by open symbols.

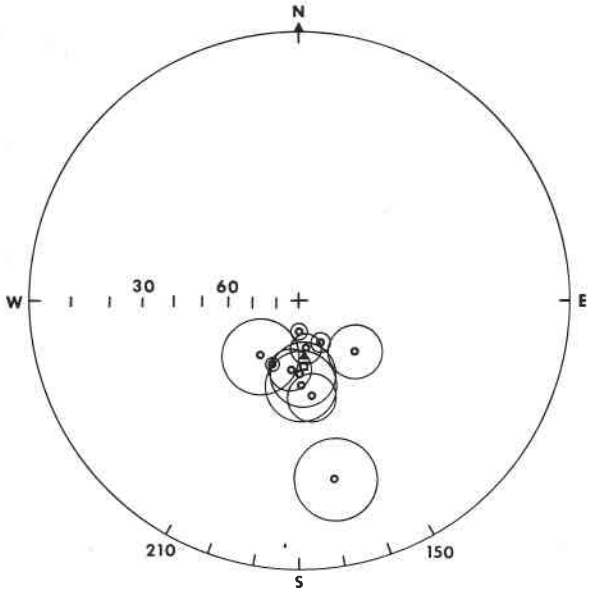


FIG. 4. Stereogram of individual site mean directions (○) with their confidence circles and the overall mean direction (□). Also shown is the reversed direction of the present axial dipole field (▲).

paleomagnetic pole not far from the average North American Triassic pole and on that basis was originally interpreted by us to indicate a Mesozoic age (Ressetar and Martin 1976). Subsequent K–Ar dating, however, has shown this dike to be about 46 Ma old and this direction has therefore been included in our mean result. The dike is slightly altered to the extent that olivine has been replaced by antigorite (Garner 1956), and secondary magnetite may have formed during the serpentinization process. Alternately, the dike may have acquired its magnetization during an excursion.

This latter possibility seems especially attractive in light of the rapid polarity reversal rate during the Eocene (Heirtzler *et al.* 1968). However, one of the more notable results of this study is that no normally magnetized units were found, although Løvlie and Opdyke (1974) found two normal andesites. Except for one possible anomalous age (site 8, 34.2 Ma), the K–Ar ages of the intrusions are in the range 42 to 46 Ma. Since during this period normal polarity is interspersed with seven short reversed intervals it would appear that intrusion was mostly restricted to one short interval.

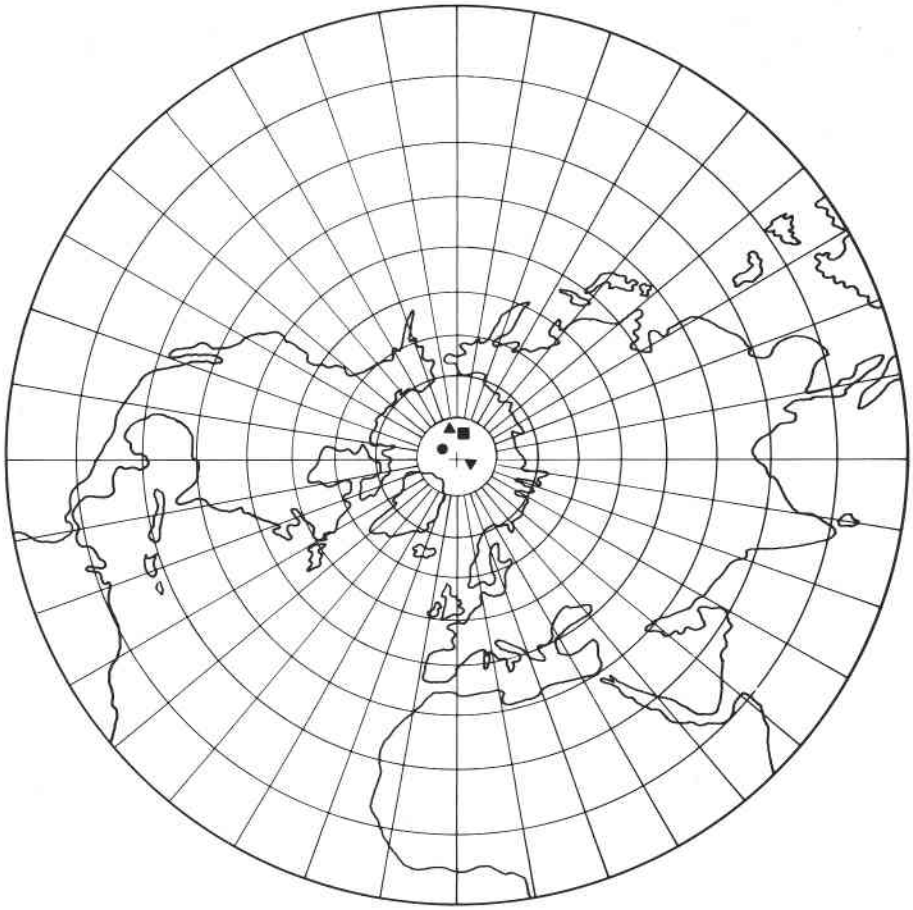


FIG. 5. Paleomagnetic pole position from this study (●) compared with other Eocene poles from tectonically stable regions of North America. References: (▼) Løvlie and Opdyke (1974), (▲) Shive *et al.* (1977), (■) Shive and Pruss (1977).

The geomagnetic pole calculated from our results is compared with other recently measured Eocene poles in Fig. 5. A discussion of secular variation by McElhinny and Merrill (1975) suggests that none of these poles individually have sufficient age or geographic span to insure that secular variation has been averaged. However, all of the poles and their average (86.5°N, 195.1°E) do not differ significantly from the present geographic pole. We therefore concur with Shive *et al.* (1977) that little apparent polar wander of North America has occurred since the Paleocene.

The cause of the Eocene activity in the Valley and Ridge is still speculative. The ages of the Valley and Ridge intrusions postdate by about 10 million years the change in direction of North Atlantic sea-floor spreading associated with the opening of the Norwegian Sea (Core *et al.* 1974) and the generation of the North Atlantic Tertiary Province. The proximity of the Eocene poles to the present geographic pole may reflect this directional change as drift of North America changed from northerly to a more westerly direction. Why this event should express itself in igneous activity in the southern Appalachians is difficult to determine, although previous workers (Zartman *et al.* 1967; Dennison and Johnson 1971) have noted that the intrusions are situated along a long lasting, deep seated zone of crustal weakness.

The recognition of olivine basalts as contemporaneous with the more felsic rocks makes Dennison and Johnson's (1971) argument for derivation from a granitic pluton appear dubious, while Hall's (1974) proposal for fractional crystallization of a basaltic magma is strengthened. The Eocene magnetic directions obtained from units of volcanic breccia and diatreme suggest a shallow depth of dike emplacement and possible volcanism.

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Pelecocrinus sp. cf. *P. banffensis* Laudon, Parks, and Spreng, from the Banff Formation (Lower Mississippian), Kananaskis Valley, Alberta, Canada

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A new occurrence of *Pelecocrinus* cf. *P. banffensis* is described from the Banff Formation of Osage age, Kananaskis Valley, Alberta.

On décrit une nouvelle occurrence de *Pelecocrinus* cf. *P. banffensis* dans la formation de Banff datant de l'Osagien, dans la vallée de Kananaskis en Alberta.

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